```
    // 检查文件是否存在
    FILE* f = fopen(img_path.c_str(), "r");
    if (f == NULL) {
        cout << "错误：文件不存在！路径：" << img_path << endl;
        return -1;
    }
    fclose(f);

    Mat img = imread(img_path, IMREAD_COLOR);
    if (img.empty()) {
        cout << "图片读取失败！可能是文件格式不支持" << endl;
        return -1;
    }
    cout << "图片读取成功！" << endl;

    // ===================== 任务2：输出图像信息 =====================
    int height = img.rows;
    int width = img.cols;
    int channels = img.channels();
    string dtype = "8U";

    cout << "\n图像基本信息" << endl;
    cout << "图像尺寸（宽×高）：" << width << " × " << height << endl;
    cout << "图像通道数：" << channels << endl;
    cout << "像素数据类型：" << dtype << endl;
```

```
图片读取成功！
```

```
    if (img.empty()) {
        cout << "图片读取失败！可能是文件格式不支持" << endl;
        return -1;
    }
    cout << "图片读取成功！" << endl;

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    cout << "\n图像基本信息" << endl;
    cout << "图像尺寸（宽×高）：" << width << " × " << height << endl;
    cout << "图像通道数：" << channels << endl;
    cout << "像素数据类型：" << dtype << endl;

    // ===================== 任务3：显示原图 =====================
    imshow("Original Image", img);
    waitKey(0);
    destroyAllWindows();

    // ===================== 任务4：转灰度图并显示 =====================
    Mat gray_img;
    cvtColor(img, gray_img, COLOR_BGR2GRAY);
```

```
图片读取成功！

图像基本信息
```

```cpp
if (img.empty()) {
    cout << "图片读取失败！可能是文件格式不支持" << endl;
    return -1;
}
cout << "图片读取成功！" << endl;

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// ===================== 任务3：显示原图 =====================
imshow("Original Image", img);
waitKey(0);
destroyAllWindows();

// ===================== 任务4：转灰度图并显示 =====================
Mat gray_img;
cvtColor(img, gray_img, COLOR_BGR2GRAY);
```
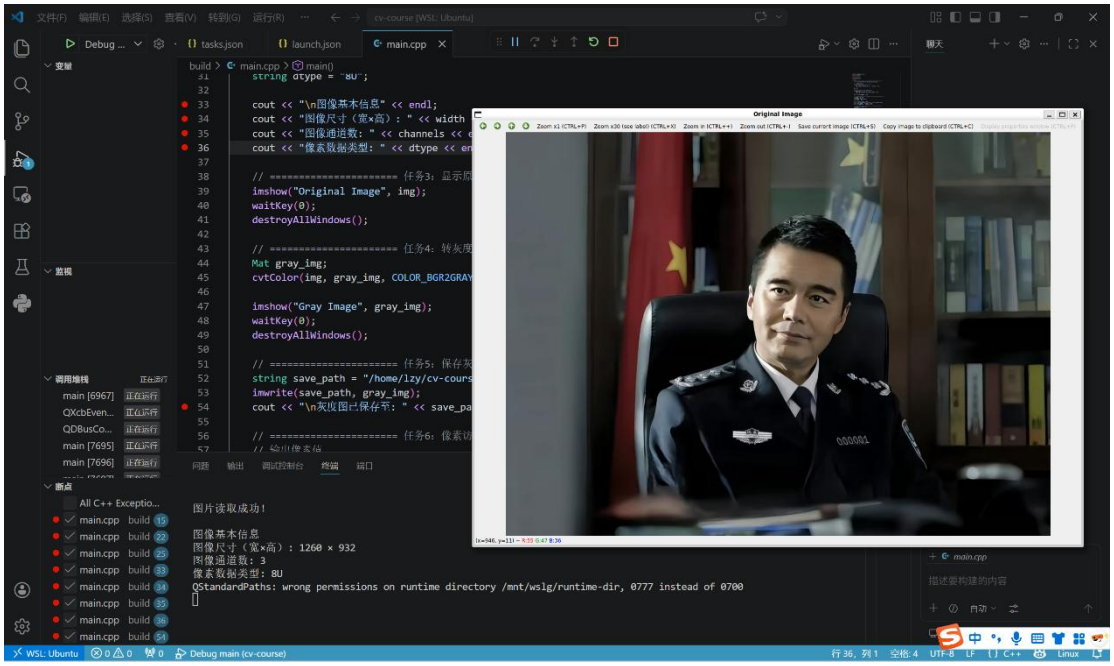
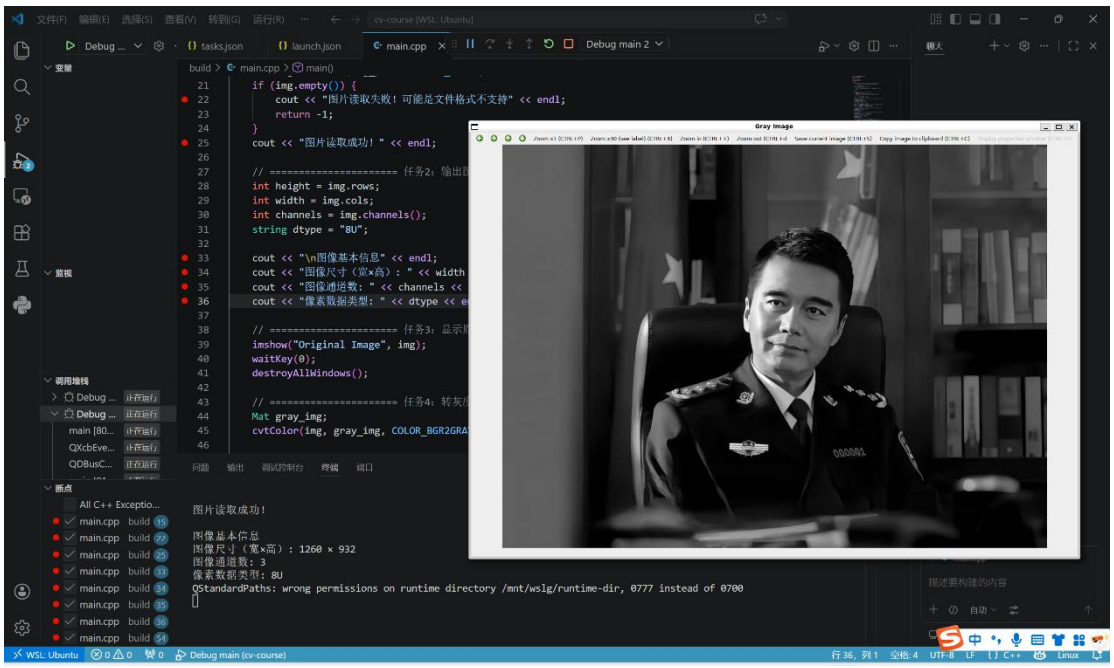

```
图片读取成功！

图像基本信息
图像尺寸（宽×高）: 1260 × 932
```

```cpp
if (img.empty()) {
    cout << "图片读取失败！可能是文件格式不支持" << endl;
    return -1;
}
cout << "图片读取成功！" << endl;

// ===================== 任务2：输出图像信息 =====================
int height = img.rows;
int width = img.cols;
int channels = img.channels();
string dtype = "8U";

cout << "\n图像基本信息" << endl;
cout << "图像尺寸（宽×高）: " << width << " × " << height << endl;
cout << "图像通道数: " << channels << endl;
cout << "像素数据类型: " << dtype << endl;

// ===================== 任务3：显示原图 =====================
imshow("Original Image", img);
waitKey(0);
destroyAllWindows();

// ===================== 任务4：转灰度图并显示 =====================
Mat gray_img;
cvtColor(img, gray_img, COLOR_BGR2GRAY);
```

```
图片读取成功！

图像基本信息
图像尺寸（宽×高）: 1260 × 932
图像通道数: 3
```


Original Image
main.cpp
imshow("Original Image", img);
waitKey(0);
destroyAllWindows();
// ========== 任务4：转灰度
Mat gray_img;
cvtColor(img, gray_img, COLOR_BGR2GRAY
imshow("Gray Image", gray_img);
waitKey(0);
destroyAllWindows();
string save_path = "/home/lzy/cv-cours
imwrite(save_path, gray_img);
cout << "\n灰度图已保存至：" << save_pa
// ========== 任务6：像素访
图片读取成功！
图像基本信息
图像尺寸（宽×高）: 1260 × 932
图像通道数: 3
像素数据类型: 8U
QStandardPaths: wrong permissions on runtime directory /mnt/wslg/runtime-dir, 0777 instead of 0700


Gray Image
if (img.empty()) {
cout << "图片读取失败！可能是文件格式不支持" << endl;
return -1;
}
cout << "图片读取成功！" << endl;
// ========== 任务2：输出
int height = img.rows;
int width = img.cols;
int channels = img.channels();
string dtype = "8U";
cout << "\n图像基本信息" << endl;
cout << "图像尺寸（宽×高）: " << width
cout << "图像通道数: " << channels <<
cout << "像素数据类型: " << dtype << e
// ========== 任务3：显示
imshow("Original Image", img);
waitKey(0);
destroyAllWindows();
// ========== 任务4：转灰
Mat gray_img;
cvtColor(img, gray_img, COLOR_BGR2GRA
图片读取成功！
图像基本信息
图像尺寸（宽×高）: 1260 × 932
图像通道数: 3
像素数据类型: 8U
QStandardPaths: wrong permissions on runtime directory /mnt/wslg/runtime-dir, 0777 instead of 0700

```
    destroyAllWindows();

    // ====================== 任务4：转灰度图并显示 ======================
    Mat gray_img;
    cvtColor(img, gray_img, COLOR_BGR2GRAY);

    imshow("Gray Image", gray_img);
    waitKey(0);
    destroyAllWindows();

    // ====================== 任务5：保存灰度图 ======================
    string save_path = "/home/lzy/cv-course/build/gray.png";
    imwrite(save_path, gray_img);
    cout << "\n灰度图已保存至：" << save_path << endl;

    // ====================== 任务6：像素访问 + 裁剪 ======================
    // 输出像素值
    if (channels == 3) {
        Vec3b pixel = img.at<Vec3b>(502, 205);
        cout << "\n坐标 (502,205) 的像素值："
             << "B=" << (int)pixel[0] << " "
             << "G=" << (int)pixel[1] << " "
             << "R=" << (int)pixel[2] << endl;
    } else {
        uchar pixel = gray_img.at<uchar>(502, 205);
        cout << "\n坐标 (502,205) 的像素值：" << (int)pi
```

```
图像尺寸（宽x高）: 1260 x 932
图像通道数: 3
像素数据类型: 8U
QStandardPaths: wrong permissions on runtime directory /mnt/wslg/runtime-dir, 0777 instead of 0700

灰度图已保存至: /home/lzy/cv-course/build/gray.png

坐标 (502,205) 的像素值: B=20 G=25 R=26

裁剪区域：左上角(400, 400)
裁剪后尺寸：宽400 × 高400
```

```
    Mat cut_img = img(Rect(0, 0, cut_w, cut_h));

    cout << "\n裁剪区域：左上角(" << cut_w << ", " << cut_h << ")" << endl;
    cout << "裁剪后尺寸：宽" << cut_img.cols << " × 高" << cut_img.rows << endl;

    imshow("Cut Image", cut_img);
    waitKey(0);
    destroyAllWindows();

    // 保存裁剪图
    string cut_save_path = "/home/lzy/cv-course/build/cut.png";
    imwrite(cut_save_path, cut_img);
    cout << "裁剪区域已保存至：" << cut_save_path << endl;

    return 0;
}
```

```
QStandardPaths: wrong permissions on runtime directory /mnt/wslg/runtime-dir, 0777 instead of 0700

灰度图已保存至: /home/lzy/cv-course/build/gray.png

坐标 (502,205) 的像素值: B=20 G=25 R=26

裁剪区域：左上角(400, 400)
裁剪后尺寸：宽400 × 高400
裁剪区域已保存至：/home/lzy/cv-course/build/cut.png
[1] + Done                       "/usr/bin/gdb" --interpreter=mi --tty=${DbgTerm} 0<"/tmp/Microsoft-MIEngine-In-xegjfyvm.2op" 1>"/tmp/Microsoft-MIEngine-Out-uh3xnpqt.itw"
(.venv-basic) lzy@lzy:~/cv-course$
```